

Table 4. Watermark schemes, within-family representative scores, and fixed operating points used in our experiments. Family-level comparisons report multiple methods per family, whereas the Hybrid uses a single pre-specified representative per family. Hyperparameters are held fixed across attacks, with no per-attack retuning.

Family	Method	Selection	Experimental Role	Mean robustness (AUROC)	Mean detectability (z -score)	Absolute score $S(m)$	Watermarking Parameters
Biased	KGW (Kirchenbauer et al., 2023)	Candidate 1	Canonical context-biased dependent sampler	0.800	8.60	0.904	$\gamma = 0.5, \delta = 2.0$, simple-1 seeding
	Unigram (Zhao et al., 2024)	Candidate 2: Hybrid representative	Fixed-green-list biased sampler	0.812	8.03	0.923	$\gamma = 0.5, \delta = 2.0$
Bias-free	DiPMark (Wu et al., 2024)	Candidate 1	Token-level bias-free reweighting comparator	0.905	3.60	1.122	$\alpha = 0.45, \gamma = 0.5$
	HeavyWater (Tsur et al., 2025)	Candidate 2	Low-entropy bias-free design	0.885	3.78	1.094	Log-normal scores, $k = 1024$, top- $p = 0.999$
	SimplexWater (Tsur et al., 2025)	Candidate 3	Low-entropy bias-free design	0.875	4.00	1.075	Simplex-code scores, $ S = m - 1$, top- $p = 0.999$
	HCW (Hu et al., 2024)	Candidate 4: Hybrid representative	Canonical unbiased reweighting design	0.917	3.12	1.160	δ -reweight variant (method=delta)
Semantic	SemStamp (Hou et al., 2024)	Candidate 1	LSH-based sentence-level semantic design	0.935	2.43	1.227	SBERT, margin 0.8, $d = 3$, $\gamma = 0.25$
	SimMark (Dabiriaghdam & Wang, 2025)	Candidate 2	Similarity-based sentence-level design	0.925	2.62	1.201	Instructor-Large, cosine range [0.68, 0.76], $K = 250$
	PMark (Huo et al., 2025)	Candidate 3: Hybrid representative	Rejection-based semantic design	0.945	2.12	1.266	Rejection parameter $\rho = 0.25$
Distribution-preserving	CGW (Christ et al., 2024)	Candidate 1: Hybrid representative	Canonical distribution-preserving sampler	0.453	-6.63	only candidate	Security parameter $\lambda = 128$; fixed secret key across runs
Training-time baseline	GaussMark (Block et al., 2025)	Outside Hybrid	Structural watermark in model weights	0.835	7.80	outside Hybrid	Gaussian perturbation scale $\sigma = 0.05$ (fixed calibration at $\alpha = 0.05$)

How the Hybrid Is Instantiated

For family-level comparison, we report multiple methods per family. For the Hybrid itself, Section 4.1 first chooses a *family* for one estimated edit regime:

1. Estimate the token edit rate $\hat{\epsilon}$ and semantic flip rate $\hat{\epsilon}_s$ from the edited LLM outputs.
2. Compute $D_{\text{req}}^{\text{tok}}$ for token-level families and $D_{\text{req}}^{\text{sem}}$ for semantic families.
3. Keep only feasible families, meaning the score must be below the corresponding stealth limit, and then choose the family with the smaller score. If a distribution-preserving method is already reliable enough, its score is 0, and it is chosen immediately.

To choose one *concrete* representative within a selected family, we use a second, fixed rule. For each candidate method m , we compute the absolute score $S(m) = \text{AUC}(m) + \frac{1}{1 + \max(\bar{z}(m), 0)}$. Here, $\text{AUC}(m)$ is the mean robustness score (AUROC) across the four main oblivious attacks in Table 3 (DIPPER, OPT-2.7B, watermark-removal prompting, and synonym substitution), so larger values indicate stronger robustness. Likewise, $\bar{z}(m)$ is the mean z -score over the same attacks. We use *keyless* to mean that this detectability score is computed by an outside observer who does not know the secret watermark key and relies only on statistical traces in the text. Smaller $\bar{z}(m)$ therefore indicates lower detectability to such an observer. Higher $S(m)$ thus favors methods that are both more robust and less keylessly detectable.

Under this rule, Unigram is chosen from the biased family, HCW from the bias-free family, and PMark from the semantic family. CGW is the only distribution-preserving candidate, and GaussMark is listed separately because it is a training-time method outside the Hybrid selector.

In the last column, γ denotes the green-list fraction, i.e., the fraction of the vocabulary treated as favored tokens, and δ the logit-bias strength, i.e., the amount by which those favored tokens are boosted, in KGW (Kirchenbauer et al., 2023) and Unigram (Zhao et al., 2024). The parameter α in DiPMark (Wu et al., 2024) controls the strength of the bias-free reweighting. In HeavyWater (Tsur et al., 2025), k is the score dimension used to construct the watermark scores. In SimplexWater (Tsur et al., 2025), $|S|$ is the size of the simplex codebook used by the method. In SemStamp (Hou et al., 2024), d is the hash dimension that determines the semantic bucket structure. In PMark (Huo et al., 2025), ρ is the rejection parameter controlling how selectively candidate sentences are accepted. In CGW (Christ et al., 2024), λ is the security parameter governing the cryptographic strength of the scheme. In GaussMark (Block et al., 2025), σ is the standard deviation of the Gaussian perturbation applied to model weights.